

Rethinking dominance in Wu tone sandhi: a hierarchical account

Evidence from Xiangshan Wu

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Tone, Stress, and Prosody Workshop: Methods for Cross-Linguistic Research

Overview

- Background: the LDS-RDS (left-dominant/right-dominant sandhi) framework
- The problem: intra-domain variation in Northern Wu
- Data: Xiangshan Wu tone sandhi
- Theoretical account: contrast preservation + hierarchical dominance
- Typological implications

1. Background: The LDS-RDS framework

The left-dominant / right-dominant dichotomy

Chinese tone sandhi dichotomy

Left-dominant sandhi (LDS)

Right-dominant sandhi (RDS)



Dominance = preservation of tones

The left-dominant / right-dominant dichotomy

Left-dominant sandhi (LDS)

Leftmost/initial tone preserved

Non-initial tones neutralised

Usually tone spreading

Canonical example:

Shanghai Wu

ts ^h ɔ	vɛ	
<i>fried</i>	<i>rice</i>	
34	13	citation tones
3	4	sandhi tones

Right-dominant sandhi (RDS)

Rightmost/final tone preserved

Non-final tones neutralised

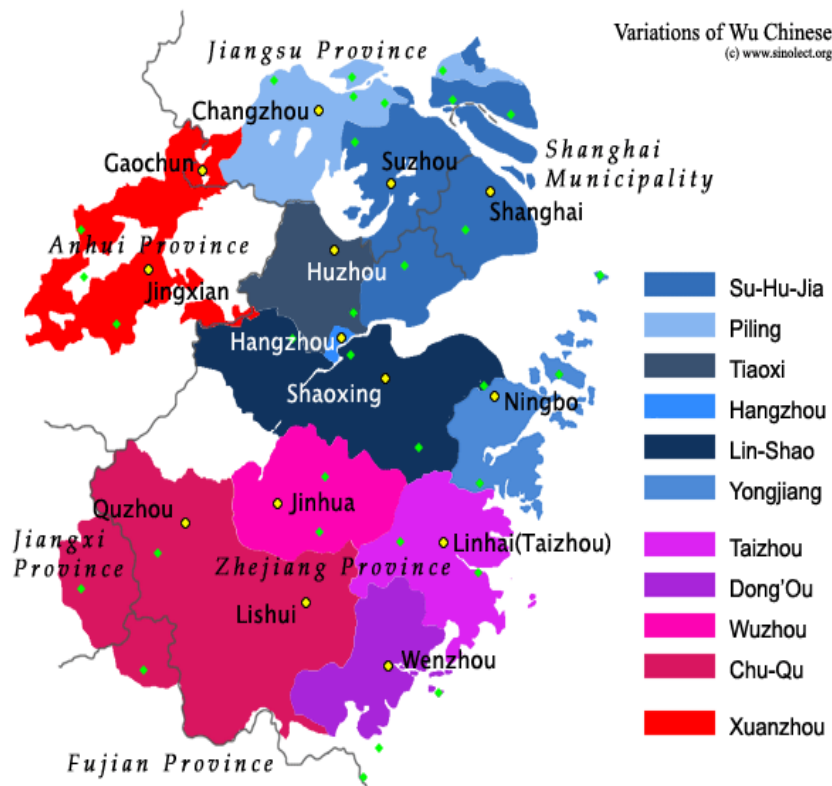
Usually tone replacement/simplification

Canonical example:

Standard Mandarin

yu	san	
<i>rain</i>	<i>umbrella</i>	
213	213	citation tones
35	213	sandhi tones

The standard typological picture of Wu tone sandhi



Macro level

- Northern Wu ≈ LDS; Shanghai
- Southern Wu ≈ RDS; Wenzhou

Within a dialect

- LDS & RDS often co-exist
- Correlated with syntax

Example: Shanghai Wu

- LDS in most structures
- RDS in Verb-Object, etc.

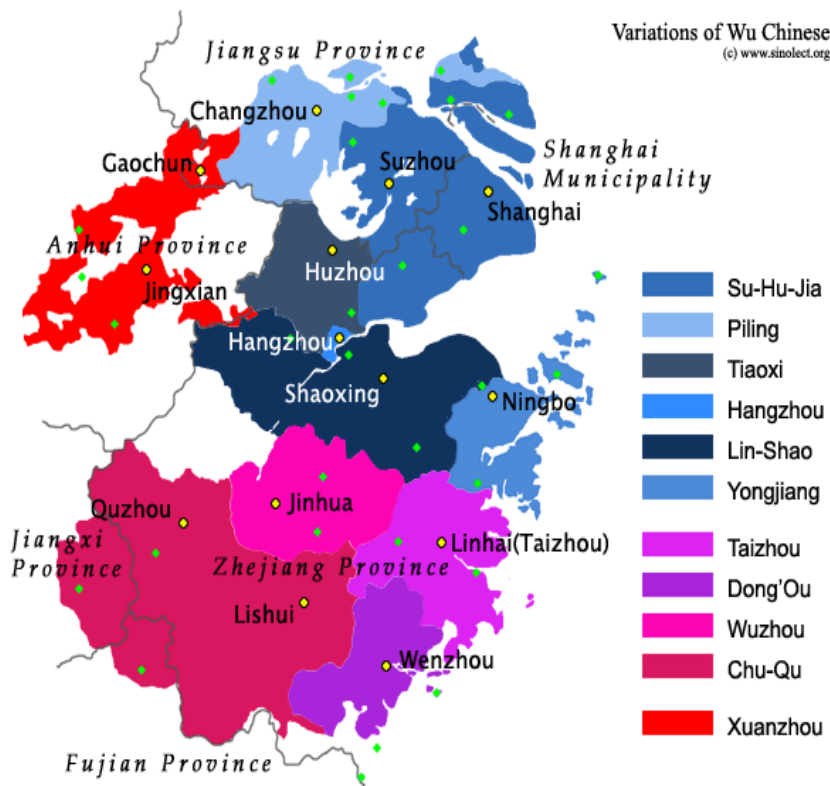
ts^hɔ
fried
34
3

ve
rice
13
4

NP: LDS

citation tones
sandhi tones

The standard typological picture of Wu tone sandhi



Macro level

- Northern Wu ≈ LDS; Shanghai
- Southern Wu ≈ RDS; Wenzhou

Within a dialect

- LDS & RDS often co-exist
- Correlated with syntax

Example: Shanghai Wu

- LDS in most structures
- RDS in Verb-Object, etc.

ts ^h ɔ	vɛ	VP: RDS
to fry	rice	
34	13	citation tones
44	13	sandhi tones

2. The problem:

Intra-domain variation in Northern Wu

The left-dominant / right-dominant dichotomy

Northern Wu tone sandhi dichotomy

Left-dominant sandhi (LDS)

Right-dominant sandhi (RDS)



Syntax-bound directionality

Citation tone preservation

Complete neutralisation in LDS

Variation within a single domain

Expected

Syntax-bound directionality

Citation tone preservation

Complete neutralisation in LDS

Reality

Syntax ↔ dominance mapping is unstable

Opacity in sandhi patterns

Incomplete neutralisation

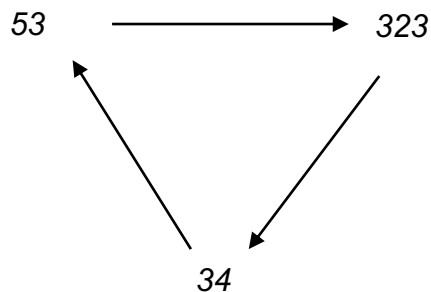
Variation within a single domain

- Syntax ↔ dominance mapping is unstable
 - Shanghai: LDS/RDS within same structure conditioned by frequency & transparency (Xu et al. 1981)

Examples	Left-dominant	Right-dominant	
拔河 /baʔ ¹³ u ¹³ / <i>tug of war</i>	[baʔ ¹¹ u ¹³]		word, closely connected
拔草 /baʔ ¹³ ts ^h ɔ ²⁴ / <i>to weed</i>	[baʔ ¹¹ ts ^h ɔ ¹³]	[baʔ ²² ts ^h ɔ ²⁴]	intermediate degree of connection
拔树 /baʔ ¹³ zɿ ¹³ / <i>to pull out a tree</i>		[baʔ ²² zɿ ¹³]	phrase, loosely connected

Variation within a single domain

- Syntax $\leftrightarrow$ dominance mapping is unstable
- Opacity in sandhi patterns
 - Wuxi: initial syllable undergoes chain shift before spreading (Chan & Ren 1988)



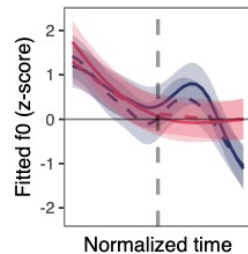
te	mɛ	
'bad luck'		
323	13	citation tones
3	4	sandhi tones

Variation within a single domain

- Syntax $\leftrightarrow$ dominance mapping is unstable
- Opacity in sandhi patterns
- Incomplete neutralisation in LDS
 - Non-initial syllables retain contour—
Shaoxing (Wen et al. 2023)

Other languages such as Jinshan (Rose & Yang 2024), etc.

A. T1TX



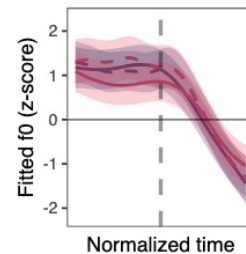
σ_2 Contour

— F
— R

σ_2 Register

— H
— L

C. T3TX



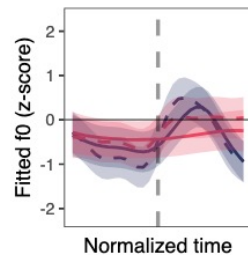
σ_2 Contour

— F
— R

σ_2 Register

— H
— L

B. T2TX



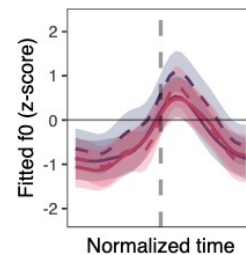
σ_2 Contour

— F
— R

σ_2 Register

— H
— L

D. T4TX



σ_2 Contour

— F
— R

σ_2 Register

— H
— L

Variation within a single domain

Syntax ↔ dominance mapping is unstable

Opacity in sandhi patterns

Incomplete neutralisation in LDS

Variation is the
default condition of
Northern Wu sandhi

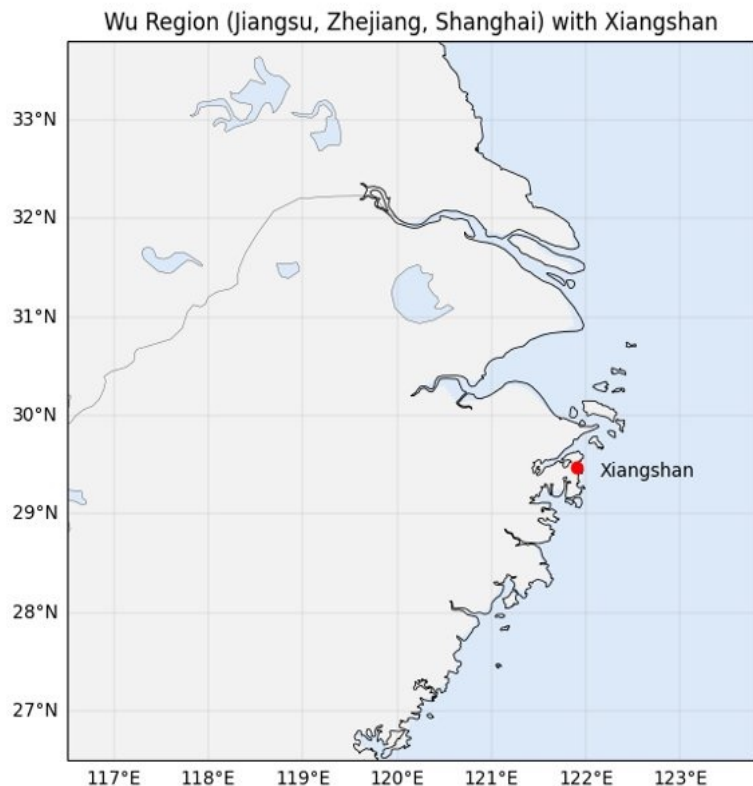
The diagnostic problem

- **Dominance diagnosis: compare sandhi contours directly with citation tones**
 - If sandhi matches initial citation tone → LDS
 - If sandhi matches final citation tone → RDS
- Conclusion for examples beforementioned: neither LDS nor RDS
 - Intra-domain variation renders the system unclassifiable under this criterion
- ⇒ Is the LDS-RDS distinction itself to blame — or the diagnostic tool?

3. Data:

Xiangshan Wu tone sandhi

Xiangshan Wu at a glance



Xiangshan (象山)

- Eastern Zhejiang coast
- A member of Northern Wu
- Under-documented; rich tonal complexity

Tone system

	<i>ping</i> level	<i>shang</i> rising	<i>qu</i> departing	<i>ru</i> entering
<i>yin</i> high	HH		HL	Hq
<i>yang</i> low	LHL		LH	LHq

Methodology

- 8 native Xiangshan speakers (4F, mean age 50)
 - Born & raised in Xiangshan; Xiangshan as primary language >75% of daily use
- Semi-randomised reading task; instructions in Xiangshan
 - Monosyllables first (citation tones), then disyllables
- 287 noun-headed disyllables; 92 verb-object disyllables
- F_0 extraction: ProsodyPro, 10 equidistant points/syllable
 - Normalised via z-scores of log-transformed f_0 (per speaker)
- Sandhi pattern identification: auditory categorisation + k -means clustering

Noun-headed structures

- Disyllabic noun-headed tone sandhi

$\sigma 1 \backslash \sigma 2$		HH <i>yinping</i> or <i>yinshang</i>	HL <i>yinqu</i>	LHL <i>yangping</i>	LH <i>yangqu</i>
HH	<i>yinping</i>	HMML MMMh			
	<i>yinshang</i>	MHHL HHHH			
LHL	<i>yangping</i>	LLHH	LHML	LLLH	
	<i>yangshang</i>	LLHH LLHL			

- **Overall LDS**
- **Initial tone:** opacity; historical-conditioned
- **Non-initial tone:** incomplete neutralisation

Verb-headed structures

- Disyllabic verb-headed tone sandhi

$\sigma_1 \backslash \sigma_2$	HH <i>yinping</i> <i>yinshang</i>	or	LH <i>yangqu</i>	LHL <i>yangping</i>	HL <i>yinqu</i>
HH <i>yinping</i>	HMML MMHH			MMHL MMHH	
HH <i>yinshang</i>					
LHL <i>yangshang</i>	LLHH		LLLH	LLRF (rise-fall)	LLHL

- Overall RDS
- Register-based LDS element

4. Theoretical account

Theoretical move 1: dominance as contrast preservation

- **Standard view: dominance = phonetic matching of sandhi to citation tone**
 - Does the sandhi output resemble the initial (LDS) or final (RDS) citation tone?
- Problem: Xiangshan noun-headed LDS
 - *yinping* & *yinshang* both surface as HH — but trigger distinct sandhi
 - Citation-matching approach: unclassifiable
- **Proposal: dominance = preservation of tonal CONTRASTS**
 - Whichever syllable's contrasts (synchronic OR historical) govern the sandhi, that syllable is dominant
 - Xiangshan noun-headed: initial syllable's HISTORICAL contrasts preserved → LDS

Theoretical move 2: hierarchical dominance

- Even correctly diagnosed domains show residual opposite-direction influence
 - Incomplete neutralisation in LDS
 - Register-based LDS in RDS

H-register: sandhi selection
L-register: neutralisation

$\sigma_2 \backslash \sigma_1$		HH <i>yinping</i> or <i>yinshang</i>	HL <i>yinqu</i>	LHL <i>yangping</i>	LH <i>yangqu</i>
HH	<i>yinping</i>	HMML MMMh			
	<i>yinshang</i>	MHHL HHHH			
LH L	<i>yangping</i>	LLHH	LHML	LLLH	
	<i>yangshang</i>	LLHH LLHL			

$\sigma_2 \backslash \sigma_1$		HH <i>yinping</i> or <i>yinshang</i>	LH <i>yangqu</i>	LHL <i>yangping</i>	HL <i>yinqu</i>
HH <i>yinping</i>	HH <i>yinshang</i>	HMML MMHH		MMHL MMHH	
LHL <i>yangshang</i>		LLHH	LLLH	LLRF (rise-fall)	LLHL

Theoretical move 2: hierarchical dominance

- **Even correctly diagnosed domains show residual opposite-direction influence**
 - Incomplete neutralisation in LDS
 - Register-based LDS in RDS
- **Proposal: dominance is hierarchically organised as a structural property**
 - One mechanism governs the domain globally
 - A second modulates specific tonal environments within it
- **Xiangshan:**
 - **Noun-headed:** LDS globally, RDS embedded
 - **Verb-headed:** RDS globally, LDS embedded

Contrast with Rose & Yang (2024)

Citation-matching approach

Compares sandhi contour directly with citation tone of each syllable

Xiangshan/Jinshan result:
'Neither LDS nor RDS'

Problem: can't diagnose systems

- where sandhi preserves historical rather than synchronic contrasts
- where sandhi shows layered dominance

Contrast-preservation approach (this paper)

Asks: whose contrasts (synchronic or historical) govern the sandhi?

Xiangshan result:
Noun-headed → LDS (historical)
Verb-object → RDS (synchronic)

Hierarchy captures residual opposite-direction modulation

5. Typological implications

Typological & diachronic implications

- **More principled classification of sandhi systems**
 - Separate global dominance direction from embedded local modulation
 - Accommodates internal variation without abandoning LDS-RDS
- **Historical-synchronic asymmetry as a diagnostic tool**
 - Which level of contrast (synchronic vs. historical) a sandhi system reflects tells us about its diachronic stage
- **Window onto the development of Northern Wu sandhi**
 - Layered dominance = historical traces of LDS-RDS interaction
- **Approaching a new Northern Wu dialect: anticipate variation, not uniformity**

Conclusion

- **Wu tone sandhi has more variation than the syntax-bound LDS-RDS dichotomy suggests**
 - Intra-domain variation is the norm across Northern Wu
- **Dominance should be defined as preservation of tonal contrasts**
 - Synchronic or historical , whichever the sandhi system reflects
- **Dominance is hierarchically organised**
 - One mechanism global at domain level; a second modulates specific tonal environments
- $\Rightarrow$ Variation is not noise; it is structured, and historically interpretable

References

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